

ANDREW AGDM, AB, Canada

WMO: 712860

Lat: 53.9176N

Lon: 112.2783W

Elev: 625

StdP: 94.04

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.7	-29.2	-36.4	0.1	-32.7	-32.7	0.2	-29.2	11.7	-0.3	10.4	-2.9	2.4	140	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	27.8	17.7	26.1	17.0	24.4	16.2	19.8	25.6	18.4	23.9	17.3	22.4	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	13.5	23.2	16.2	12.4	21.3	15.2	11.6	19.9	59.4	25.8	54.7	24.2	51.0	22.2	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	9.1	8.0	-37.3	31.0	3.7	1.5	-39.9	32.1	-42.1	33.0	-44.1	33.8	-46.8	34.9
			-37.3	22.5	3.7	1.4	-40.0	23.4	-42.2	24.2	-44.2	25.0	-46.9	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-12.5	-12.5	-6.1	3.2	10.3	14.3	16.4	14.9	10.2	3.3	-5.9	-12.8	(6)
	DBStd	12.37	8.32	7.78	7.30	5.36	4.44	2.95	2.74	3.16	4.34	4.75	6.85	7.92	(7)
	HDD10.0	3541	697	630	499	209	51	4	0	3	49	213	478	708	(8)
	HDD18.3	5995	955	863	757	454	251	123	73	113	244	467	728	966	(9)
	CDD10.0	611	0	0	0	5	59	134	198	154	56	4	0	0	(10)
	CDD18.3	24	0	0	0	0	1	3	13	6	1	0	0	0	(11)
	CDH23.3	597	0	0	0	1	59	86	220	156	74	2	0	0	(12)
	CDH26.7	109	0	0	0	0	11	10	47	26	16	0	0	0	(13)
(14) Wind	WSAvg	3.7	3.6	3.5	3.9	4.3	4.3	3.9	3.1	3.0	3.5	3.8	3.6	3.4	(14)
(15) Precipitation	PrecAvg	413	19	11	17	30	44	70	81	54	33	19	19	16	(15)
	PrecMax	574	69	23	37	65	118	128	139	103	82	44	77	39	(16)
	PrecMin	221	3	1	6	2	7	21	30	11	4	1	1	3	(17)
	PrecStd	73	15	6	8	16	22	28	27	19	21	11	16	9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	5.4	10.4	22.4	28.3	28.1	30.5	29.2	28.9	22.1	11.6	5.8	(19)
		MCWB	2.3	2.2	5.8	11.0	13.7	16.7	20.2	18.3	16.6	12.3	6.6	2.6	(20)
	2%	DB	3.7	3.0	6.6	18.5	24.8	25.8	27.7	27.0	25.5	16.5	7.2	3.1	(21)
		MCWB	1.4	0.8	3.4	8.9	12.4	15.8	19.8	18.0	14.9	9.5	3.8	0.8	(22)
	5%	DB	1.8	1.0	4.6	15.0	22.3	23.7	25.7	25.0	22.1	13.5	4.7	1.0	(23)
		MCWB	0.1	-0.5	2.0	7.4	11.5	14.9	18.5	17.2	13.3	7.6	2.2	-0.8	(24)
	10%	DB	-0.3	-0.9	3.0	12.1	19.9	21.7	23.8	23.1	18.9	10.9	2.5	-1.5	(25)
		MCWB	-1.7	-2.2	1.0	5.8	10.4	14.1	17.4	16.6	11.9	6.3	0.8	-2.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.6	6.3	11.4	14.8	18.8	22.2	20.7	17.0	12.5	6.9	3.0	(27)
		MCDB	4.9	4.8	9.9	20.5	23.8	24.9	27.7	26.2	26.8	21.2	11.0	5.9	(28)
	2%	WB	1.6	1.0	3.6	9.3	13.4	17.3	20.5	19.1	15.4	10.2	4.0	1.0	(29)
		MCDB	3.5	2.6	6.2	17.7	22.0	22.9	26.4	24.8	23.6	15.6	6.8	2.9	(30)
	5%	WB	0.3	-0.4	2.2	7.7	12.3	16.1	19.1	17.8	14.0	8.3	2.3	-0.8	(31)
		MCDB	1.9	1.1	4.4	14.0	20.4	21.8	24.6	23.4	21.3	12.5	4.3	0.9	(32)
	10%	WB	-1.7	-2.3	1.1	6.3	11.2	15.0	17.9	16.8	12.6	6.8	0.8	-2.7	(33)
		MCDB	-0.4	-1.0	2.8	11.3	18.2	20.1	23.0	22.1	17.9	10.4	2.4	-1.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.1	10.2	11.4	13.7	11.9	12.6	13.6	13.7	11.5	9.3	9.7	(35)
		MCDBR	11.4	13.0	11.2	17.4	17.9	15.8	16.2	17.2	19.8	16.4	11.8	11.9	(36)
		MCWBR	9.4	11.0	8.4	9.2	7.8	7.5	9.1	9.3	9.9	9.7	8.4	10.0	(37)
	5% WB	MCDBR	11.2	12.6	10.7	16.0	15.9	13.5	15.0	15.6	18.1	15.5	10.9	11.8	(38)
		MCWBR	9.4	10.8	8.1	8.9	7.2	7.1	8.7	8.9	9.3	9.5	8.1	10.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.254	0.264	0.306	0.333	0.346	0.360	0.360	0.370	0.309	0.279	0.252	0.241	(40)
	taud		2.294	2.331	2.305	2.360	2.378	2.360	2.396	2.330	2.495	2.553	2.446	2.276	(41)
	Ebn at Noon		728	850	876	898	901	888	881	847	871	834	744	665	(42)
	Edh at Noon		56	76	98	107	112	115	109	110	82	61	48	46	(43)
(44) All-Sky Solar Radiation	RadAvg		0.92	1.96	3.49	4.67	5.72	5.97	6.08	4.99	3.46	1.97	0.93	0.67	(44)
	RadStd		0.10	0.15	0.26	0.41	0.44	0.40	0.31	0.36	0.36	0.26	0.10	0.06	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page